



Improving the economic efficiency of small and medium-sized cities in the Yangtze River delta: What role for borrowed size?

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ARTICLE INFO

Keywords:

Borrowed size
Small and medium-sized cities
Economic efficiency
Yangtze River Delta

ABSTRACT

Research on urban agglomerations has increased in recent decades, while research on small and medium-sized cities in urban agglomerations is less and under-theorized. The paper takes the Yangtze River Delta as an example and explores the role of borrowed size in improving the economic efficiency of small and medium-sized cities in urban agglomerations. The results show that borrowed size has a significant positive effect on the economic efficiency of small and medium-sized cities. Furthermore, improving resource misallocation is an important mechanism for promoting economic efficiency in small and medium-sized cities through borrowed size. This paper provides multidimensional empirical evidence for promoting coordinated development of urban agglomerations.

1. Introduction

Urban agglomeration, as a highly developed spatial form of integrated cities, has become one of the most important carriers for global economic development (Batten, 1995; Duranton & Puga, 2004; Fang & Yu, 2017). However, in most urban agglomerations, small and medium-sized cities are facing development difficulties such as a single industrial structure (Liu et al., 2024), lagging infrastructure construction (Elena & Maria, 2018), and population loss (Escudero-Gómez et al., 2023). Their improvement in economic efficiency is severely constrained, resulting in a significant gap in the economic and social development level of central cities (Cuberes et al., 2021; Yang & Yang, 2024). At the same time, small and medium-sized cities may also play an important role, for example, as specialists in certain products or commodities. In recent decades, research on urban agglomerations has increased, while research on small and medium-sized cities in urban agglomerations is less and under-theorized (Bell & Jayne, 2009; Fahmi et al., 2014). Therefore, it is of vital importance to study the way to improve the economic efficiency of small and medium-sized cities in urban agglomerations.

The Yangtze River Delta, as the most dynamic urban cluster in China, has seen some of its subordinate small and medium-sized cities, such as Jiangyin, Zhangjiagang, and Changshu, surpassing some provincial capital cities in terms of the total economic scale. These cities have developed high-end manufacturing systems by leveraging the developed

productive service industry in central cities like Shanghai. While these cities have achieved rapid development through borrowed size, not all small and medium-sized cities in the country have met their economic growth goals. Some have even fallen into the shadow of agglomeration without achieving substantial growth. Therefore, this paper aims to explore the path to improve the economic efficiency of small and medium-sized cities in the Yangtze River Delta from the perspective of borrowed size, providing references for the economic development of small and medium-sized cities worldwide.

The concept of borrowed size was first proposed by Alonso (1973). He observed that smaller cities within the Atlantic coastal agglomerations had significantly higher incomes compared to independently developed small cities of the same scale. These cities not only retained the advantageous characteristics of small-scale cities but also reaped the benefits of larger-scale cities through borrowed size. Meijers and Burger (2017) categorized borrowed size into two dimensions: borrowed performance and borrowed functions. They argued that the similar characteristics mentioned by Alonso (1973) about small and medium-sized cities and central cities can refer to typical urban functions, including artifacts, activities, and facilities, as well as performance levels equivalent to those of central cities. Additionally, Camagni et al. (2016) emphasized the need to differentiate the role of borrowed functions within the concept of borrowed size. The traditional concept of borrowed size has failed to distinguish the borrowed activities of small and

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<https://doi.org/10.1016/j.cities.2025.105954>

Received 6 September 2024; Received in revised form 16 March 2025; Accepted 1 April 2025

Available online 11 April 2025

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medium-sized cities in terms of population scale and urban functions from those of central cities. Based on this, the paper investigates the impact of borrowed size on the economic efficiency of small and medium-sized cities about both borrowed performance and borrowed functions.

The existing literature on borrowed size can be broadly categorized into two groups. The first group employs the concept of borrowed size to elucidate the spatial distribution of economic activities, encompassing the economic progress of peripheral cities or suburbs (Hesse, 2016; Phelps, 1998) and the establishment of businesses in proximity to metropolitan areas (Polèse & Shearmur, 2006). This line of research is at its nascent stage, primarily relying on descriptive observations and case studies to furnish objective evidence for confirming the presence of borrowed size effects. The second group emphasizes the influencing factors of borrowed size, such as urban functions (Burger et al., 2015), urban size (Volgmann & Rusche, 2020), geographical proximity (Malý, 2016), and other related aspects. However, there remains a dearth of literature that explores the role of borrowed size in intercity factor allocation, particularly from the perspective of small and medium-sized cities. Therefore, this paper examines the impact of borrowed size on the economic efficiency of small and medium-sized cities in the Yangtze River Delta.

The main contributions of this paper are as follows. First, previous studies focused on the spillover effects between cities of different scales. However, there is a limited exploration of the mechanisms behind intercity spatial spillovers. This paper adopts a perspective of borrowed size to investigate its impact on the economic efficiency of small and medium-sized cities. It provides a comprehensive analysis of the intercity factor allocation mechanisms, considering both performance and function dimensions. Second, this paper analyzes the heterogeneity of the metropolitan area, traffic acceleration, G60 science, and innovation corridor, and further verifies the mechanism of borrowed size from the perspective of improving labor mismatch and capital mismatch. The research results provide new empirical evidence for the development of small and medium-sized cities.

The remainder of this paper is organized as follows. Section 2 addresses the theoretical background and research hypothesis; Section 3 describes the methodology and data; Section 4 discusses the empirical results; Section 5 presents the main conclusions and discussions, and provides policy implications.

2. Theoretical background and research hypothesis

2.1. Theoretical background

The concept of “Borrowed Size” is proposed by American economist Alonso (1973): “The concept of a system of cities has many facets, but one of particular interest is the concept of borrowed size, whereby a small city or metropolitan area exhibits some of the characteristics of a larger one if it is near other population concentrations.....small metropolises in megapolitan complexes, such as that on the Atlantic seaboard, have much higher incomes than independent metropolises of equivalent size.” He further explains that these small cities not only retain the advantages of being small, such as reduced congestion but also enjoy the benefits of being part of a larger area through borrowed size. This means that residents of small cities can utilize the shopping and entertainment facilities of larger cities, while businesses can share facilities like warehousing and business services. Additionally, the labor market in these small cities benefits from a wider and more flexible range of supply and demand. Using the concept of borrowed size, Alonso (1973) points out that when looking at the scale and growth of a city, we should not look at the city in isolation, but also consider its neighbors.

Although Alonso (1973) does not provide a clear definition of borrowed size, the concept can be derived from his description: small and medium-sized cities can benefit from the agglomeration economies of nearby central cities while avoiding the costs associated with

agglomeration, thus exhibiting similar characteristics to central cities (Burger et al., 2015). Moreover, Meijer and Burger (2015) contend that these similar characteristics with central cities can encompass both the urban functions of central cities and an equivalent level of performance. Accordingly, they divide the concept of borrowed size into two dimensions of borrowed performance and borrowed functions and explain: “There are performance and functional dimensions of borrowing size. As one city builds the support base of a nearby, linked city that consequently maintains a higher level of functionality not normally found in a city of its size (borrowed functions), the support lending city simultaneously has access to these functions and to other agglomeration benefits, allowing it to perform better (borrowed performance) than when positioned in isolation.” Based on this, this paper divides borrowed size into two dimensions: borrowed performance and borrowed functions. By borrowed functions, small and medium-sized cities maintain a higher level of functions that are usually absent in cities of the same size, and by borrowed performance, they can benefit from the agglomeration benefits of central cities, making them perform better than when they are positioned in isolation.

2.2. Research hypothesis

2.2.1. The effect of borrowed size on economic efficiency of small and medium-sized cities

Interaction among cities is a crucial prerequisite for effective borrowed performance (Meijers & Burger, 2017; Phelps, 2004). With the growth of networked urban agglomerations, small and medium-sized cities located near central cities can leverage the considerable market opportunities presented by the central cities, thereby stimulating market demand for small and medium-sized cities. To be specific, first, small and medium-sized cities tend to borrow the market demand of central cities. Mowery & Rosenberg (1979) discovered that the bidirectional interaction between market demand and innovation is a key factor in technological progress and put forward the theory of “technological innovation and demand interaction.” Technological progress is driven by market interests and is influenced by both market supply and demand (Romer, 1990). Therefore, it can be observed that technological innovation is a kind of market behavior, and the expansion of market demand can promote technological progress in small and medium-sized cities, ultimately enhancing economic efficiency. Second, small and medium-sized cities can borrow from the labor market of central cities. The increasing costs of land and living in central cities may result in labor relocation from central cities to small and medium-sized cities (Xing & Zhang, 2017). The diversified labor supply of small and medium-sized cities improves the labor-position matching degree and improves labor productivity. The upgrading of human capital in small and medium-sized cities can promote the concentration of urban resources in high-efficiency industries, leading to improved economic efficiency.

Small and medium-sized cities located near central cities are no longer limited by their size. Instead, they can tap into various functions that are unique to central cities. In other words, these smaller cities can benefit from economies of scale without experiencing negative externalities such as congestion caused by excessive concentration. By borrowing functions from central cities, small and medium-sized cities can enjoy high-level functions, thereby creating a more favorable development environment. For instance, small and medium-sized cities can diversify their financing channels by borrowing financial service functions from central cities. They can also address issues of information asymmetry by borrowing information service functions, consequently making use of technology diffusion and information sharing from central cities. Furthermore, by borrowing research service functions, small and medium-sized cities can accelerate their learning and absorption of new knowledge and technologies, facilitate the dissemination and exchange of knowledge, and enhance their level of technological innovation. Lastly, borrowed commercial service functions allow small and

medium-sized cities to improve their operational efficiency, and management levels, and reduce operating costs.

Accordingly, we put forth the following hypothesis:

Hypothesis 1. Borrowed size can promote the economic efficiency of small and medium-sized cities from the aspects of borrowed performance and borrowed functions.

2.2.2. Mechanism analysis

Previous studies have indicated that resource misallocation is a key factor leading to a decline in urban economic efficiency, and enhancing resource allocation efficiency is crucial for improving economic efficiency (Hsieh & Klenow, 2009; Oberfield, 2013). By borrowing size, small and medium-sized cities undertake the industrial transfer of central cities and get more capital, technology, management experience, etc. They improve the level of capital mismatch between cities. In addition, the central city has diversified industrial types and wider demand for the labor force, which can provide jobs with more matching skills for workers in the surrounding small and medium-sized cities. With the network development of urban agglomerations and the reduction of inter-city transportation costs, small and medium-sized cities can improve the degree of labor force matching by borrowed size, reduce the friction between skilled labor and jobs, and improve the degree of labor mismatch. This paper aims to further analyze how borrowed size can address the issue of resource misallocation in small and medium-sized cities.

The polycentric structure of urban agglomerations enables small and medium-sized cities to achieve “borrowed size” and enhance economic efficiency. The essence of polycentric configuration lies in the organic integration of multiple sub-centers within an urban agglomeration, facilitating efficient resource utilization and coordinated development through functional specialization, improved transportation infrastructure, and advancements in information technology (Boussaïw et al., 2018). First, small and medium-sized cities prioritize identifying and selecting potential sub-centers with strong economic foundations and development potential while maintaining geographical proximity (Pan et al., 2024). These cities also adopt rational functional specialization based on resource endowments and industrial structures, complementing the functions of sub-centers and avoiding homogeneous competition (Wang et al., 2019). Second, robust transportation infrastructure is a prerequisite for polycentric urban configuration. Small and medium-sized cities enhance the free and efficient flow of resources via modern transportation networks, including highways and railways, creating a city network characterized by functional complementarity and resource sharing. This foundation allows them to leverage the performance of central cities. Efficient public transportation systems and information and communication technologies further connect sub-centers, reducing commuting time and costs while improving operational efficiency (Hu et al., 2018; Yang et al., 2019). Third, polycentricity fosters complementarity and synergy among cities within an urban agglomeration (Wu et al., 2021). In a polycentric structure, small and medium-sized cities, guided by different sub-centers, develop specialized service sectors and functional zones according to their unique strengths, generating a borrowing effect (Wang et al., 2022). Polycentric configuration not only facilitates resource sharing and improves the efficiency of public service facilities but also disperses economic risks, ensuring regional economic stability.

The development of new towns represents a key form of urbanization for small and medium-sized cities and a vital driver of regional economic and social progress. In modern urban agglomerations, new towns help optimize resource allocation, achieve borrowed size, and improve economic efficiency (Xue et al., 2013). As a contemporary urban form, new towns feature diverse functions, comprehensive infrastructure, and pleasant ecological environments, enhancing the complementarity between small and large cities while addressing challenges like land scarcity and environmental pollution in larger cities (Chen et al., 2023;

Van Leynseele & Bontje, 2019). First, improvements in transportation infrastructure enhance connectivity between small and large cities, promoting the flow of talent, capital, and technology throughout the urban agglomeration. This fosters resource-sharing networks, allowing small and medium-sized cities to leverage the performance of central cities and boost economic efficiency (García-López & Muñiz, 2010; Wang et al., 2018). Second, scientific planning and green development of new towns provide livable environments that attract populations and businesses from large cities, alleviating urban pressures (Shen et al., 2020). Additionally, the efficient allocation and sharing of resources in new towns enhance intercity resource flows, introducing high-quality public services such as higher education, healthcare, and cultural amenities, thereby strengthening the attractiveness and competitiveness of small and medium-sized cities (Robinson et al., 2021). Third, the development of new towns enhances regional innovation capacity and economic competitiveness by establishing technology parks and business incubators, attracting high-end talent and innovation resources to drive economic transformation (Governa & Sampieri, 2020). The borrowed functions effects of new town development support the rational layout of production, living, and ecological spaces while advancing regional innovation and competitiveness.

Under the neoliberal planning framework, small and medium-sized cities within urban agglomerations can optimize resource allocation and enhance economic efficiency through borrowed size. Neoliberal urban planning emphasizes market-driven approaches, advocating reduced government intervention and resource allocation through market mechanisms (Zhou et al., 2019). In this context, small and medium-sized cities collaborate with larger cities or other regional centers to leverage scale advantages, fostering economic, social, and spatial structure optimization. First, labor and capital mobility are pivotal mechanisms in the neoliberal model. Within urban agglomerations, especially between small and large cities, labor and capital flows provide technological, knowledge, and financial resources that enhance labor market flexibility. This process enables performance borrowing effects, mitigating social inequality and reducing resource waste caused by spatial mismatches (Zhen et al., 2019). Second, small and medium-sized cities benefit from the scale effects of large cities through spatial agglomeration and functional complementarity. These cities often demonstrate strong industrial specialization and complementary functions. By implementing strategic spatial planning and functional positioning, they integrate into the industrial and supply chains of large cities, advancing local service industry upgrading, realizing borrowed functions effects, and improving economic efficiency. This neoliberal planning model not only boosts economic efficiency and social equity within urban agglomerations but also enhances the competitiveness and sustainability of small and medium-sized cities, providing critical support to address the challenges of rapid urbanization.

Therefore, we propose the following hypothesis:

Hypothesis 2. Borrowed size can improve the resource misallocation of small and medium-sized cities through the multi-centrality allocation of urban agglomerations, the development of new towns, and the market mechanism under the framework of neoliberal planning.

3. Methodology and data

3.1. Empirical strategy

Referring to Wooldridge (2016) and Sierra (2020), this paper adopts the fixed effects model to examine the effect of borrowed size on the economic efficiency of small and medium-sized cities. The basic estimation model is set as follows:

$$DTFP_{it} = \beta_0 + \beta_1 \text{borrow_size}_{it} + \beta_2 X_{it} + u_i + v_i + \varepsilon_{it} \quad (1)$$

where i represents the region and t represents the year. The dependent variable $DTFP_{it}$ signifies the urban economic efficiency of a particular

region i in a given year t . The core explanatory variable $borr_size_{it}$ represents the borrowed performance or borrowed functions of the region i in that respective year t . Additionally, a set of control variables denoted by β_0 is the intercept term, β_2 stands as the collection of regression coefficients for the control variables. Furthermore, μ_t and ν_i indicate the time-fixed effects and individual-fixed effects, respectively. ε_{it} represents the error term. The main focus of this section lies on the parameter β_1 . A significantly positive value of β_1 indicates that borrowed performance or borrowed functions can enhance the economic efficiency of small and medium-sized cities, thus validating the theoretical hypothesis mentioned above. Conversely, a significantly negative value of β_1 suggests a negative impact of borrowed performance or borrowed functions on the economic efficiency of small and medium-sized cities, thereby implying the existence of agglomeration shadow effects.

3.2. Variable selection

3.2.1. Independent variable: Borrowed size

3.2.1.1. Dimension of performance. Alonso (1973) suggests that population potential measures the accessibility from a given location to other population centers, indicating the opportunity for interaction with people from different places. It can serve as a measure of borrowed performance. Camagni and Capello (2015) define borrowed performance as the accessibility of secondary cities to the population of larger urban areas and formally propose a method for measuring borrowed performance. Referring to these studies, borrowed performance (viz. the accessibility of the small and medium-sized cities to the performance of the central cities) is calculated as the population of the central cities discounted by distance from each small and medium-sized city. The calculation formula is as follows:

$$borr_perf_i = \sum_{m=1}^n \frac{pop_m}{w_{i,m}}, \forall i \neq m \quad (2)$$

where i represents the small and medium-sized cities, m represents the central city. pop represents the vector of the central city population. w is an $n \times n$ distance weight matrix between central cities and small and medium-sized cities. Each entry in the matrix is equal to 0 if the two cities are central cities or small and medium-sized cities, and equal to the inverse of the geographical distance in kilometers if the two cities are cities one of central and one of small and medium-sized.

3.2.1.2. Dimension of functions. Camagni et al. (2016) proposed a methodology for measuring the borrowed functions. Expanding upon the research of Duranton and Puga (2002) on the measurement of urban functional specialization, they utilized the proportion of employment in advanced occupations as a metric for assessing the urban high-level functions. Liu and Chen (2017) measured urban high-level functions by the ratio of advanced service industries to productive manufacturing industries, considering that data on the number of people employed in advanced occupations in China is currently unavailable. Drawing on the research of Duranton and Puga (2002), Camagni et al. (2016), and Liu and Chen (2017), the paper adopts the proportion of individuals employed in advanced service industries as an indicator for gauging the urban high-level functions. Concerning the definition of high-level service industries, Yuan and Chen (2011) delineated the current high-level service industries in China into four categories: business services, information services, R&D services, and financial services. Building on their research about the “National Economic Industry Classification (GB/T 4754-2002),” selected four major industry categories, namely “information transmission, computer services and software industry,” “financial industry,” “leasing and business services,” and “scientific research, technical services and geological exploration industry,” to represent high-level service industries. Therefore, in reference to the

research of Xuan et al. (2019), this paper formulates the equation for measuring the urban high-level functions as follows:

$$func_m = \sum_{k=1}^4 \frac{func_{m,k}}{l_m} \quad (3)$$

when $k = 1, 2, 3, 4$, $func_{m,k}$ respectively denotes the urban high-level functions of city i in the information transmission, computer services and software industry, financial industry, leasing and business services, and scientific research, technical services and geological exploration industry. l represents the scale of employees in all industries in the city.

Similarly to the calculation method used for assessing borrowed performance, borrowed functions (viz. the accessibility of the small and medium-sized cities to high-level functions of the central cities) are calculated as the high-level functions of the central cities discounted by distance from each small and medium-sized city:

$$borr_func_i = \sum_{m=1}^n \frac{func_m}{w_{i,m}}, \forall i \neq m \quad (4)$$

where $func$ signifies the vector of high-level functions of the central city, and w is the same distance matrix as applied for the “borrowed performance” indicator.

3.2.2. Dependent variable: Economic efficiency

The paper assesses urban economic efficiency by employing the measure of total factor productivity (TFP). Drawing from the approach developed by Coe and Helpman (1995), the Solow residual method is used for estimating TFP. The calculation formula for TFP is as follows:

$$TFP_{it} = Y_{it} / (K_{it}^\alpha L_{it}^{1-\alpha}) \quad (5)$$

where Y as the output level, K as the capital stock, L as the labor input, and as the capital-output elasticity. The selection of indicators and their data explanations are as follows: (1) The output level (Y) is measured by the actual GDP of each city. To account for price factors, the nominal GDP is adjusted to real GDP using the year 2001 as the base period. (2) The labor input (L) is represented by the employment figures in each city. (3) The capital stock (K) is estimated using the perpetual inventory method, specifically adopting the approach proposed by Shan (2008). The calculation formula is $K_{it} = K_{i,t-1}(1 - \delta_t) + I_{it}/p_t$, in which K represents the capital stock, δ represents the depreciation rate, I represents fixed asset investment, and p represents the fixed asset investment price index calculated based on the year 2001 as the base period. (4) The capital-output elasticity (α) is derived through regression analysis, referencing the research conducted by Jiang and Feng (2012).

3.2.3. Control variables

Referring to existing research, this paper selects the following control variables:

- (1) **lnGDP**: denoting the level of economic development, where the per capita nominal GDP is adjusted to per capita real GDP based on the year 2002 while also applying a logarithmic transformation. GDP per capita reflects the level of wealth created under limited conditions and serves as the foundation for improving economic efficiency (Chen et al., 2024).
- (2) **Indus**: representing the industrial structure, is measured by the proportion of value added from the secondary and tertiary sectors of GDP. The industrial structure determines how resources are allocated and the productivity of production. A well-structured industry can effectively enhance labor productivity and capital efficiency, and promote innovation and technological progress, thus improving overall economic efficiency (Rasekhi et al., 2017).

- (3) *InWage*: signifies the wage level and is determined by the average wage of employed individuals, adjusted to real wages using the consumer price index for residents, and expressed in logarithmic form. Wage levels have a significant impact on economic efficiency (Malul et al., 2021). Higher wages are typically associated with a more skilled and productive workforce, which leads to greater innovation, increased productivity, and improved efficiency in economic processes.
- (4) *FDI*: denotes Foreign Direct Investment and is calculated as the proportion of actual foreign capital utilization to GDP. Foreign Direct Investment (FDI) not only brings capital but also introduces technology, management experience, and access to global markets. These factors enhance production processes and expand market opportunities, ultimately boosting the city's economic efficiency (Guo et al., 2022).
- (5) *Invest*: indicates fixed asset investment, which reflects the proportion of total fixed asset investment amount about GDP. Fixed asset investments, including infrastructure, real estate, and industrial facilities, can enhance production capacity and modernize capital, thus improving economic efficiency (Halkos & Tzeremes, 2009).
- (6) *Gov*: represents government intervention and is measured by the proportion of local fiscal expenditure to GDP, serving as an indicator of the degree of government intervention. Government intervention, through policies and regulations, can create a favorable economic environment by addressing market failures, promoting innovation, and ensuring equitable resource distribution, all of which contribute to enhancing economic efficiency (Gao & Gao, 2024).
- (7) *InTrans*: signifies transportation infrastructure and is determined by the logarithm of road network density within built-up areas. Efficient transportation systems reduce transaction costs, facilitate the movement of goods and labor, and expand market access (Melo et al., 2013).
- (8) *InInfor*: represents information infrastructure, expressed as the logarithm of the number of internet users per ten thousand people. Strong information infrastructure enhances communication capabilities, supports digitalization, and fosters innovation, all of which positively impact productivity and economic efficiency (Sirgy et al., 2012).

3.2.4. Mechanism variables

The paper draws on the research of Bai and Liu (2018) and uses the following formulas to measure the Capital Mismatch Index ($MisK_{it}$) and Labor Mismatch Index ($MisL_{it}$):

$$MisK_{it} = \left(\frac{K_{it}}{K} \right) / \left(\frac{s_{it}\beta_{K_{it}}}{\beta_K} \right) - 1 \quad (7)$$

$$MisL_{it} = \left(\frac{L_{it}}{L} \right) / \left(\frac{s_{it}\beta_{L_{it}}}{\beta_L} \right) - 1 \quad (8)$$

where K_{it}/K , L_{it}/L represents the proportion of capital and labor utilized in the region i relative to the total amount of capital and labor, respectively. $s_{it}\beta_{K_{it}}/\beta_K$ ($s_{it}\beta_{L_{it}}/\beta_L$) denotes the ideal proportion for efficient capital (labor) allocation and $MisK_{it}$ ($MisL_{it}$) reflects the extent of capital (labor) mismatch in the region i . If $MisK_{it}$ ($MisL_{it}$) is greater than 0, it implies that the cost of capital (labor) utilization in the region i is lower compared to the entire economy, thereby suggesting an excessive allocation of capital (labor) in that specific region. Conversely, if $MisK_{it}$ ($MisL_{it}$) is less than 0, it implies that the actual allocation of capital (labor) in the region is lower than the optimal allocation, indicating an insufficient use of capital (labor) in that particular region.

Regarding the capital output elasticity (β_K) and labor output elasticity (β_L), the paper adopts the Solow residual method based on the research of Zhao et al. (2006) for measurement. Due to the potential

differences in urban technological levels, there may be significant variations in factor output elasticities among cities. Drawing on the research of Bai and Liu (2018), the paper employs a variable coefficient panel model to estimate the factor output elasticities for each city. Based on this, the study calculates $MisK_{it}$ and $MisL_{it}$ for each city. Considering the existence of both resource under allocation ($MisK_{it} < 0$, $MisL_{it} < 0$) and resource overallocation ($MisK_{it} > 0$, $MisL_{it} > 0$), in order to maintain consistent regression directions, the study takes the absolute values of the capital mismatch index ($MisK_{it}$) and labor mismatch index ($MisL_{it}$), and the larger the absolute values after taking the absolute values, the more severe the resource mismatch situation in the region.

3.3. Data sources and descriptive statistics

The paper adopts county-level and above cities as the geographical research units and utilizes a sample of 82 small and medium-sized cities in the Yangtze River Delta from 2003 to 2020, considering the availability and consistency of data. The relevant data required for calculating the independent variables are sourced from the *China City Statistical Yearbook*. The distance data between central cities and small-medium cities are obtained by organizing the *China Terrain Database*, using Euclidean straight-line distance. The relevant data needed for calculating the dependent variable, the control variables, and the mechanism variables are derived from the *China City Statistical Yearbook* and the *China County Statistical Yearbook*.

To define the central cities and small and medium-sized cities, the paper draws upon the research conducted by Li et al. (2019) and Wang et al. (2021). The central cities are comprised of directly governed municipalities, provincial capitals, and independently planned cities. In the case of the Yangtze River Delta urban cluster, Shanghai, Nanjing, Hangzhou, Hefei, and Ningbo are identified as the central cities. Following the *Notice of the State Council on the Adjustment of Urban Scale Division Standards* (No.51 Document of State Council in [2014]), the paper utilizes data from the initial year of 2003 to define cities with a permanent urban population of less than 1 million as small and medium-sized cities. The data is sourced from the *China Urban Construction Statistical Yearbook*. Based on this, the paper focuses on a total of 82 small and medium-sized cities, including 33 prefecture-level city districts and 49 county-level cities in the Yangtze River Delta region. The measurement methods and descriptive statistics of each variable are presented in Table 1.

4. Empirical results

4.1. Baseline regression

Table 2 reports the baseline results. In terms of the efficiency dimension, we control for year-fixed and city-fixed effects in column (1). It can be found that the coefficient of borrowed size (*Borr_size*) is significantly positive. Moreover, we all control variables in column (2), and can find that the borrowed size (*Borr_size*) remains significantly positive at the 1 % level, suggesting that borrowed performance significantly contributes to enhancing the economic efficiency of small and medium-sized cities in the Yangtze River Delta. The results indicate that small and medium-sized cities improve their economic efficiency by borrowing production factors and expanding production scale through central cities. This approach leads to the generation of agglomeration effects while circumventing the agglomeration costs associated with congestion in central cities. Consequently, borrowed strategies are conducive to improving the economic efficiency of small and medium-sized cities.

Column (3) and column (4) provide regression results in the functional dimension. We control for year and city fixed effects in column (3), finding the coefficient of borrowed size (*Borr_size*) is significantly positive. Furthermore, all control variables are included in column (4), and the estimates of borrowed size (*Borr_size*) are significantly positive

Table 1

Measurement of variables and descriptive statistics.

Variables		Measurement	Observation	Mean	Std	Min	Max
Dependent variable	Economic efficiency(<i>DTFP</i>)	Solow Complementary method	1476	11.266	0.646	9.428	13.134
Independent variable	Borrowed size (<i>Borr_size</i>)	Performance dimension: $\sum_{m=1}^n \frac{pop_m}{w_{i,m}}$	1476	17.650	9.682	5.468	66.030
		Function dimension: $\sum_{m=1}^n \frac{func_m}{w_{i,m}}$	1476	0.353	0.159	0.120	1.079
Control variable	Economic development(<i>lnPGDP</i>)	Logarithm of GDP per capita	1476	10.400	0.751	7.824	12.280
	Industrial structure(<i>Indus</i>)	The proportion of secondary and tertiary industries in GDP	1476	0.915	0.069	0.597	0.996
	Wage(<i>lnWage</i>)	The logarithm of the actual average wage of employed employee	1476	10.400	0.691	8.866	14.430
	Foreign direct investment(<i>FDI</i>)	The proportion of regional actual utilization of foreign capital to GDP	1476	0.014	0.039	0.000	0.973
	Fixed investment(<i>Invest</i>)	The proportion of fixed investment to GDP	1476	11.260	0.988	7.650	14.370
	Government intervention(<i>Gov</i>)	The proportion of government intervention to GDP	1476	0.123	0.074	0.000	1.107
	Transportation infrastructure (<i>lnTrans</i>)	Logarithm of network density in built-up area	1476	2.123	0.397	0.667	3.671
	Information infrastructure (<i>lnInfor</i>)	Logarithm of the number of users per 10,000 people accessing the Internet	1476	8.647	0.999	3.648	11.570

Table 2

Baseline regression results.

Variable	(1) Performance	(2) Performance	(3) Functions	(4) Functions
<i>Borr_size</i>	0.024*** (0.007)	0.027*** (0.007)	0.570* (0.289)	0.568** (0.246)
<i>lnPGDP</i>		0.426*** (0.059)		0.401*** (0.062)
<i>Indus</i>		−1.002 (0.923)		−0.994 (0.923)
<i>lnWage</i>		0.020 (0.018)		0.012 (0.018)
<i>FDI</i>		0.169 (0.186)		0.221 (0.201)
<i>Invest</i>		−0.110*** (0.041)		−0.119*** (0.044)
<i>Gov</i>		−0.181 (0.369)		−0.208 (0.404)
<i>lnTrans</i>		−0.087** (0.035)		−0.098*** (0.035)
<i>lnInfor</i>		0.043 (0.058)		0.045 (0.058)
Constant	10.347*** (0.101)	6.835*** (0.553)	10.523*** (0.086)	7.388*** (0.524)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
R ²	0.783	0.847	0.779	0.841
Samples	1476	1476	1476	1476

Note: (1) Robust standard error in parentheses; (2) *, ** and *** represent 10 %, 5 % and 1 % significance levels respectively.

at the 5 % level, indicating that the effect of borrowed functions leads to improving the economic efficiency of small and medium-sized cities. This result may be attributed to the fact that small and medium-sized cities benefit from accessing diversified services and intermediate product supply by borrowing advanced functions from central cities. In addition, the local labor force benefits from the expanded labor market demand, further facilitating the improvement of economic efficiency in small and medium-sized cities. Thus, the findings of the paper support the hypothesis that borrowed size influences the economic efficiency of small and medium-sized cities, indicating a significant promoting effect of borrowed size on overall economic efficiency.

4.2. Robustness test

To verify the reliability of the regression results mentioned above, the paper conducts a robustness test from four perspectives: changing the measurement method for economic efficiency in small and medium-sized cities, addressing endogeneity, removing outliers, and considering

the lagged effect of borrowed performance.

4.2.1. Changing the measurement method

To examine the robustness of the benchmark regression results, the paper calculates the economic efficiency of small and medium-sized cities in the Yangtze River Delta using two different methods: Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA). It also re-evaluates the relationship between borrowed size and economic efficiency in small and medium-sized cities. The regression results are presented in Table 3. Columns (1) and (2) display the regression results of city economic efficiency based on the DEA-Malmquist index method, while columns (3) and (4) represent the results based on the SFA method. The results consistently demonstrate that the coefficient of borrowed size is significantly positive in both performance and functional dimensions, indicating that borrowed size indeed has a significant enhancing effect on the economic efficiency of small and medium-sized cities - in line with the benchmark regression results. This confirms that

Table 3

Robustness test: changing the measurement method.

Variable	DEA-Malmquist Index		Stochastic frontier analysis (SFA)	
	(1) Performance	(2) Functions	(3) Performance	(4) Functions
<i>Borr_size</i>	0.004*** (0.001)	0.084** (0.038)	0.003*** (0.001)	0.066* (0.037)
<i>lnPGDP</i>	0.019* (0.010)	0.015 (0.010)	0.021* (0.012)	0.018* (0.010)
<i>Indus</i>	−0.104 (0.086)	−0.104 (0.088)	0.308** (0.131)	0.308*** (0.100)
<i>lnWage</i>	0.004 (0.004)	0.003 (0.004)	0.016** (0.006)	0.015*** (0.003)
<i>FDI</i>	0.050 (0.056)	0.059 (0.059)	−0.136*** (0.024)	−0.129*** (0.048)
<i>Invest</i>	−0.021** (0.009)	−0.022** (0.009)	0.008* (0.005)	0.007* (0.004)
<i>Gov</i>	−0.025 (0.065)	−0.030 (0.072)	−0.043 (0.043)	−0.047 (0.058)
<i>lnTrans</i>	−0.009 (0.007)	−0.011* (0.007)	0.007 (0.008)	0.006 (0.008)
<i>lnInfor</i>	−0.003 (0.005)	−0.003 (0.005)	−0.006 (0.007)	−0.006 (0.004)
Constant	0.861*** (0.084)	0.955*** (0.079)	−0.622*** (0.094)	−0.592*** (0.085)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
R ²	0.615	0.598	0.163	0.321
Samples	1476	1476	1476	1476

Note: (1) Robust standard error in parentheses; (2) *, ** and *** represent 10 %, 5 % and 1 % significance levels respectively.

the empirical findings of the paper are not influenced by the measurement method used for city economic efficiency, thus ensuring the robustness of the regression results.

4.2.2. Processing endogeneity

During the examination of the impact of borrowed size on the economic efficiency of small and medium-sized cities, the bidirectional causal relationship between them may lead to endogeneity issues. The previous regression results have confirmed that borrowed size positively influences economic efficiency in both performance and functional dimensions. However, it should be noted that the economic efficiency of small and medium-sized cities may also affect the borrowed size. Additionally, despite including a comprehensive set of control variables to reduce potential omitted variable bias that could affect the economic efficiency of cities, there are still numerous factors that influence city economic efficiency. Some of these factors are challenging to measure accurately due to data availability limitations. Consequently, the paper employs the following dynamic panel data model to robustly examine the relationship:

$$DTFP_{it} = \beta_0 + \alpha DTFP_{it-1} + \beta_1 \text{borrow_size}_{it} + \beta_2 X_{it} + \mu_i + \nu_t + \varepsilon_{it} \quad (30)$$

The regression analysis in the paper is conducted using the System Generalized Method of Moments (GMM). The results of the regression are presented in columns (1) and (2) of Table 4. Autocorrelation tests for AR(1) and AR(2) show that the disturbance term exhibits first-order serial correlation, but no second-order serial correlation. The Hansen test indicates that the instrumental variables are valid, with *p*-values greater than 0.1. Thus, the results of the dynamic model regression in columns (1) and (2) can be considered reliable. The findings reveal that the regression coefficients of borrowed size are significantly positive in both performance and functional dimensions. Furthermore, they pass the significance test at a 5 % level, indicating that borrowed size can indeed promote the improvement of economic efficiency in small and medium-sized cities. These results are consistent with the benchmark regression conclusion.

4.2.3. Removing outliers

The influence of outliers was mitigated by conducting a 1 % two-sided winsorization on all variables to ensure reliable estimation results. The regression results, presented in columns (3) and (4) of Table 4, reveal a positive and statistically significant regression coefficient for the borrowed size (*Borrow_size*) at the 1 % significance level. This finding implies that the impact of outliers is successfully addressed and underscores the robustness of the borrowed size's role in enhancing

economic efficiency in small and medium-sized cities.

4.2.4. Considering the lagged effect

To account for potential time lag effects in the influence of borrowed size on the economic efficiency of small and medium-sized cities, the core explanatory variable in The paper was replaced with the first lagged variable of borrowed size (*L.Borrow_size*). Regression estimation was performed using two-way fixed effects, and the results are presented in columns (5) and (6) of Table 4. The coefficient of the first lagged variable of borrowed size (*L.Borrow_size*) is significantly positive, reaffirming the robust finding that borrowed size plays a vital role in promoting the enhancement of economic efficiency in small and medium-sized cities.

4.3. Heterogeneity analysis

4.3.1. The impact of metropolitan areas

Metropolitan areas are important regulatory units in the national governance system. They experience frequent flows of factors, which facilitate network connectivity and cross-regional cooperation among cities, making them crucial for regional coordinated development. While metropolitan areas can promote the borrowed of scale from small and medium-sized cities to central cities, they can also exacerbate the suction effect of central cities on small and medium-sized cities. To examine the heterogeneous impact of metropolitan areas on the borrowed size effect, the paper divides the samples into two groups: cities within metropolitan areas and cities outside metropolitan areas. Separate regression analyses are then conducted on the subsamples.

Table 5 shows that the impact of borrowed size on urban economic efficiency is significantly positive in metropolitan cities, both in terms of performance dimension and functional dimension. However, in non-metropolitan cities, the impact of borrowed size on urban economic efficiency does not pass the test of statistical significance, but all of them pass the between-group difference coefficient test. This indicates that metropolitan areas facilitate the manifestation of the borrowed size effect in small and medium-sized cities. The main reason is that compared with the development of a single city, a metropolitan area can realize inter-city coordination and cross-regional cooperation, forming a fully functional development space, which is conducive to small and medium-sized cities learning from the scale of central cities. Specifically, the construction of metropolitan areas facilitates the free flow and rational allocation of resource factors between cities, thereby enhancing the borrowed performance of small and medium-sized cities to central cities. Additionally, the coordinated planning of metropolitan areas promotes the mutual synergy of functions among internal cities and plays a

Table 4
Robustness test: Endogeneity, outliers and lagged effect.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	Performance	Functions	Performance	Functions	Performance	Functions
<i>Borrow_size</i>	0.004** (0.002)	0.189** (0.081)	0.025*** (0.003)	0.393*** (0.121)		
<i>L.DTFP</i>	0.927*** (0.044)	0.926*** (0.036)				
<i>L.Borrow_size</i>					0.025*** (0.003)	0.751** (0.132)
AR(1)-P Value	0.009	0.007				
AR(2)-P Value	0.468	0.500				
Hansen-P Value	0.499	0.837				
Control Variables	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
R ²			0.964	0.963	0.964	0.963
Sample	1394	1394	1476	1476	1394	1394

Note:1) Robust standard error in parentheses;

2) *, ** and *** represent 10 %, 5 % and 1 % significance levels respectively;

3) The columns (1) and (2) show the endogeneity test results, the columns (3) and (4) show the outliers test results, and the columns (5) and (6) show the lagged effect test results.

Table 5

Heterogeneity test: The impact of metropolitan areas.

Variable	Performance		Functions	
	(1)	(2)	(3)	(4)
	Metropolitan cities	Non-metropolitan cities	Metropolitan cities	Non-metropolitan cities
<i>Borr_size</i>	0.035*** (0.009)	−0.040 (0.041)	0.733*** (0.263)	−1.100 (0.768)
<i>lnPGDP</i>	0.453*** (0.083)	0.409*** (0.091)	0.394*** (0.089)	0.406*** (0.090)
<i>Indus</i>	0.026 (0.720)	−1.821 (1.317)	0.147 (0.724)	−1.794 (1.285)
<i>lnWage</i>	0.014 (0.016)	0.065 (0.081)	−0.003 (0.014)	0.069 (0.080)
<i>FDI</i>	0.173 (0.182)	−1.765 (1.304)	0.229 (0.213)	−1.731 (1.298)
<i>Invest</i>	−0.091*** (0.020)	−0.379*** (0.103)	−0.104*** (0.026)	−0.371*** (0.100)
<i>Gov</i>	−0.214 (0.340)	−0.637 (0.394)	−0.212 (0.387)	−0.635 (0.416)
<i>lnTrans</i>	−0.061 (0.043)	−0.035 (0.036)	−0.093** (0.044)	−0.033 (0.038)
<i>lnInfor</i>	−0.049 (0.029)	0.135 (0.080)	−0.039 (0.031)	0.135* (0.079)
Constant	6.153*** (0.708)	7.404*** (0.841)	7.158*** (0.699)	7.210*** (0.797)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
R ²	0.899	0.835	0.889	0.835
Samples	900	576	900	576
Between-group difference	p-value = 0.000***		p-value = 0.001***	

Note: (1) Robust standard error in parentheses; (2) *, ** and *** represent 10 %, 5 % and 1 % significance levels respectively. (3) The p value of coefficient difference between groups was obtained through 1000 self-sampling using Fisher test method.

positive role in the borrowed functions of small and medium-sized cities.

4.3.2. The impact of speeding up transportation

In promoting the development of urban agglomerations, transportation planning plays a crucial role. A well-developed transportation network can accelerate the flow of production factors such as labor, capital, and technology between regions. This, in turn, facilitates the development of new markets for businesses and expands regional trade openness (Démurger, 2001; Lakshmana, 2011). The speed of transportation not only directly impacts regions but also influences the intensity of connections and interactions between cities (He et al., 2023; Shao et al., 2017). Therefore, the paper measures the speed of transportation based on whether a city has opened a high-speed rail. In the baseline regression model, we introduce the interaction term of borrowed size and whether a city has opened a high-speed rail to examine the heterogeneous effect of transportation speed on the borrowed size.

In column (1) of Table 6, the coefficient of the interaction term is significantly positive, indicating that transportation speed has a positive effect on borrowed performance. This is likely because faster transportation reduces the distance spatially and temporally, between small and medium-sized cities and central cities, allowing them to engage in transactions in larger markets at lower costs. As a result, the borrowed performance of small and medium-sized cities is promoted. Additionally, faster transportation facilitates convenient and frequent exchange and communication between small and medium-sized cities and central cities, enabling businesses in small and medium-sized cities to gain more development opportunities and learn advanced technologies and management experiences. By accessing the labor market of central cities, knowledge spillover effects are generated, thereby improving economic efficiency.

In column (2) of Table 6, the coefficient of the interaction term is not significant, indicating that transportation speed has not significantly affected the borrowed functional effect in small and medium-sized cities. This may be because although faster transportation improves the accessibility of advanced functions of central cities for small and medium-sized cities, it also enhances the attractiveness of central cities,

Table 6

Heterogeneity test: traffic acceleration and science and innovation corridor.

Variable	Traffic acceleration		The G60 Science and Innovation Corridor	
	(1)	(2)	(3)	(4)
	Performance	Functions	Performance	Functions
<i>Borr_size</i>	0.019*** (0.007)	0.441* (0.252)	0.035*** (0.005)	0.538*** (0.139)
<i>Borr_size*HSR</i>	0.005** (0.002)	0.118 (0.180)		
<i>HSR</i>	−0.198*** (0.048)	−0.151*** (0.056)		
<i>Borr_size*SIC</i>			−0.010** (0.004)	0.058 (0.133)
<i>SIC</i>			−0.950*** (0.099)	−1.335*** (0.087)
<i>lnPGDP</i>	0.410*** (0.059)	0.382*** (0.061)	0.398*** (0.042)	0.406*** (0.043)
<i>Indus</i>	−1.203 (0.914)	−1.155 (0.904)	−0.842** (0.369)	−1.017*** (0.369)
<i>lnWage</i>	0.033* (0.019)	0.024 (0.017)	0.018** (0.007)	0.012* (0.007)
<i>FDI</i>	0.172 (0.169)	0.221 (0.185)	0.186 (0.130)	0.213 (0.139)
<i>Invest</i>	−0.104*** (0.035)	−0.114*** (0.040)	−0.109*** (0.019)	−0.119*** (0.018)
<i>Gov</i>	−0.131 (0.372)	−0.161 (0.413)	−0.164 (0.179)	−0.212 (0.195)
<i>lnTrans</i>	0.043 (0.055)	0.043 (0.055)	0.043** (0.017)	0.045** (0.018)
<i>lnInfor</i>	−0.078** (0.032)	−0.090*** (0.033)	−0.087*** (0.019)	−0.098*** (0.019)
Constant	7.133*** (0.531)	7.625*** (0.503)	7.555*** (0.279)	8.205*** (0.285)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
R ²	0.855	0.848	0.962	0.961
Sample	1476	1476	1476	1476

Note: (1) Robust standard error in parentheses; (2) *, ** and *** represent 10 %, 5 % and 1 % significance levels respectively.

leading to the outflow of production factor resources from small and medium-sized cities to central cities. This outflow is detrimental to the improvement of economic efficiency in small and medium-sized cities. Therefore, the impact of transportation speed on the borrowed functional effect is twofold, resulting in no significant effect.

4.3.3. The G60 science and innovation corridor in the Yangtze River Delta

As an important part of the national strategy for the integration of the Yangtze River Delta, the G60 Science and Innovation Corridor is becoming a source of regional technological innovation and high-quality development. Each city within the Science and Innovation Corridor has its own strengths and advantages, and they complement each other. Industry alliances such as new materials (Jinhua), intelligent driving (Suzhou), artificial intelligence (Songjiang), biomedicine (Hangzhou), and the general aviation industry (Wuhu) have been established successively within the corridor. In addition, industry cooperation demonstration parks have been established in multiple cities. The G60 Science and Innovation Corridor, as the new core of innovation and creativity in the Yangtze River Delta, has practical significance in exploring the heterogeneous effects of borrowed size on promoting regional collaborative innovation and development. Based on this, the interaction term of borrowed size and whether a city has joined the G60 Science and Innovation Corridor is introduced in the baseline regression model. The estimation results are shown in column (3) and column (4) of Table 6. From the perspective of performance dimension, the coefficient of the interaction term in column (3) is significantly negative, indicating that joining the G60 Science and Innovation Corridor weakens the borrowed performance effect of small and medium-sized cities, forming a “concentration shadow”. From the perspective of functional dimension, the coefficient of the interaction term in column (4) does not pass the significance test, indicating that joining the G60 Science and Innovation Corridor has not significantly affected the borrowed functional effect of small and medium-sized cities. The reason may be that the G60 Science and Innovation Corridor is still in the early stage of development, and the regional collaborative mechanism has not yet matured, resulting in the central cities within the corridor still exerting a suction effect on small and medium-sized cities, with a weak driving effect. This leads to the concentration shadow effect in small and medium-sized cities within the Science and Innovation Corridor.

4.4. Mechanism test

It is widely acknowledged that resource mismatches limit the economic efficiency of cities (Hsieh & Klenow, 2009; Oberfield, 2013). If borrowed size can alleviate resource mismatches, it suggests that improving borrowed size can enhance the economic efficiency of small and medium-sized cities by addressing these mismatches.

The estimated results are presented in Table 7. Based on the findings of columns (1) and (2), it can be observed that the coefficient of the borrowed size (*Borr_size*) variable is significantly negative. This indicates that borrowed size significantly reduces the degree of capital misallocation in small and medium-sized cities, improving both performance and functional dimensions. The borrowed size effect stimulates the flow of capital from central cities to small and medium-sized cities, attracting more enterprises to invest and establish factories in the latter, and facilitating the transfer of industries from central cities. Consequently, the capital allocation efficiency in small and medium-sized cities is enhanced, leading to a reduction in intercity capital misallocation. Additionally, based on the results of columns (3) and (4), the coefficient of the borrowed size (*Borr_size*) variable is also significantly negative, demonstrating that the borrowed size has a pronounced impact on ameliorating labor misallocation in small and medium-sized cities. By borrowing a diverse range of labor resources from central cities, small and medium-sized cities can enhance labor matching, alleviate friction between skilled labor and job positions, boost labor productivity, and address labor misallocation.

Table 7

Mechanism test.

Variable	Capital mismatch		Labor mismatch	
	(1) Performance	(2) Functions	(3) Performance	(4) Functions
<i>Borr_size</i>	−0.011*** (0.002)	−0.174** (0.086)	−0.009*** (0.003)	−0.196* (0.115)
<i>lnPGDP</i>	−0.025 (0.025)	−0.015 (0.025)	−0.313*** (0.032)	−0.304*** (0.032)
<i>Indus</i>	1.285*** (0.223)	1.299*** (0.225)	−0.865*** (0.262)	−0.868*** (0.261)
<i>lnWage</i>	0.017** (0.008)	0.021** (0.008)	−0.042*** (0.010)	−0.039*** (0.010)
<i>FDI</i>	−0.732*** (0.182)	−0.761*** (0.185)	−0.088 (0.144)	−0.106 (0.146)
<i>Invest</i>	0.297*** (0.033)	0.301*** (0.034)	−0.005 (0.014)	−0.002 (0.014)
<i>Gov</i>	−0.659*** (0.185)	−0.640*** (0.195)	0.434*** (0.113)	0.443*** (0.116)
<i>lnTrans</i>	0.081*** (0.016)	0.086*** (0.016)	0.047** (0.020)	0.051*** (0.019)
<i>lnInfor</i>	−0.056*** (0.016)	−0.056*** (0.016)	−0.000 (0.012)	−0.001 (0.012)
Constant	−0.350* (0.206)	−0.675*** (0.188)	6.553*** (0.299)	6.320*** (0.283)
City FE	YES	YES	YES	YES
Year FE	YES	YES	YES	YES
R ²	0.888	0.886	0.961	0.961
Sample	1476	1476	1476	1476

Note: (1) Robust standard error in parentheses; (2) *, ** and *** represent 10 %, 5 % and 1 % significance levels respectively.

In conclusion, the improvement of both capital misallocation and labor misallocation are crucial mechanisms through which the borrowed size promotes the efficiency enhancement of small and medium-sized city economies. This is achieved through two main aspects. Firstly, with the development of urban agglomeration networks, small and medium-sized cities located adjacent to central cities can expand their market demands by leveraging the sizable markets of central cities. This allows for expansion of production scale, activation of agglomeration effects, and improvement of production efficiency, ultimately enhancing local resource allocation efficiency. Secondly, small and medium-sized cities adjacent to central cities can easily access intermediate products and service facilities produced by central cities, facilitating functional synergy between cities. Small and medium-sized cities can concentrate on developing industries based on their resource endowments, establishing a specialized division of labor between central cities and small and medium-sized cities. These efforts further enhance resource allocation efficiency. Overall, these findings highlight the significance of the borrowed size in facilitating the economic efficiency improvement of small and medium-sized cities, by tackling capital and labor misallocation issues.

5. Conclusions and discussions

5.1. Conclusions

Against the backdrop of promoting high-quality development and striving for the goal of achieving shared prosperity, the paper specifically focuses on the developmental challenges faced by small and medium-sized cities. Its objective is to explore strategies to enhance their economic efficiency and address issues of regional imbalances and insufficiency. By utilizing a sample of small and medium-sized cities at the county level or higher within the Yangtze River Delta urban agglomeration, this paper empirically investigates the influence of borrowed size on the economic efficiency of such cities from both performance and functionality perspectives.

The findings indicate the following: (1) Irrespective of performance or functionality dimensions, borrowed size plays a facilitating role in

improving the economic efficiency of small and medium-sized cities in the Yangtze River Delta urban agglomeration. This outcome remains robust even after adjusting for alternate measures of the explained variable and the core explanatory variable, addressing endogeneity concerns, eliminating outliers, and considering time lag effects. (2) Analysis of heterogeneity reveals that the impact of borrowed size on enhancing economic efficiency is more pronounced in urban agglomeration cities compared to non-urban agglomeration cities. The acceleration of transportation infrastructure facilitates the effectiveness of borrowed performance effects in small and medium-sized cities, while the presence of the G60 Science and Technology Innovation Corridor weakens such effects. However, the influence of both factors on borrowed functional effects is not considered significant. (3) Mechanism examinations demonstrate that improving capital misallocation and labor misallocation serve as crucial mechanisms through which borrowed size contributes to the enhancement of economic efficiency in small and medium-sized cities, irrespective of performance or functionality dimensions.

5.2. Discussions

First, this paper empirically examines the impact of borrowing scale on the economic efficiency of small- and medium-sized cities and its underlying mechanisms. The results provide empirical evidence supporting the theoretical proposition that borrowing scale promotes the economic development of small- and medium-sized cities within urban agglomerations, reinforcing confidence in China's urban agglomeration planning and development strategies. Furthermore, these findings validate the macro-level argument in the existing literature that small- and medium-sized cities can leverage the functions and economic externalities of larger cities to improve their development performance and enhance the functional division of labor and network connectivity within urban agglomerations (Meijers et al., 2016; Volkmann & Rusche, 2020). The results also align with studies showing that borrowing scale fosters economic growth, strengthens spatial spillover effects, and optimizes the spatial structure of urban agglomerations (Gong & Zhong, 2021; Chen et al., 2024; Cai et al., 2023). Building on these insights, this paper further decomposes borrowing scale into borrowing performance and borrowing functions to analyze its specific effects on the economic efficiency of small- and medium-sized cities, offering a more refined and detailed perspective for policy and development strategies.

Second, the heterogeneity analysis reveals that within urban agglomerations, the effect of borrowing scale on enhancing the economic efficiency of small- and medium-sized cities is more pronounced. This finding underscores the effectiveness of government efforts in urban agglomeration planning and highlights the importance of regional-level strategies for urban agglomeration and metropolitan planning to promote coordinated development among large, medium, and small cities. Furthermore, transportation infrastructure enhances borrowing performance for small- and medium-sized cities, whereas technological innovation corridors weaken this effect. This suggests that transportation infrastructure facilitates the free flow of resources between cities (Deng, 2013), creating favorable conditions for borrowing performance. Conversely, technological innovation resources tend to concentrate in larger cities, potentially forming an agglomeration shadow (Bernstein, 2009) that limits the borrowing performance of small- and medium-sized cities. This paper provides practical policy recommendations for urban agglomeration planning and coordinated development while also identifying existing challenges within urban agglomerations.

Finally, this paper investigates the mechanisms by which borrowing scale affects the economic efficiency of small- and medium-sized cities. The findings align with the consensus that factor mobility and optimized resource allocation enhance urban economic efficiency (Gao et al., 2020; Kim et al., 2014; Zhou et al., 2024) and offer valuable empirical evidence on how reducing capital and labor misallocation improves the economic efficiency of these cities (Li et al., 2013; McGowan & Andrews,

2015; Wu et al., 2017). Moreover, this paper deepens existing studies on borrowing scale, urban specialization, and factor mobility by addressing the more practical issues of capital and labor misallocation.

5.3. Policy recommendations

First, strengthen policy support to enhance the borrowed size effect. The results indicate that borrowed size has a significant positive impact on improving the economic efficiency of small and medium-sized cities. On one hand, fiscal policies should be optimized to promote cross-regional industrial collaboration. Governments should formulate fiscal support policies tailored to small and medium-sized cities, such as providing special industrial support funds and fiscal subsidies, to encourage enterprises to strengthen industrial cooperation with core cities. On the other hand, regional industrial cooperation and information-sharing platforms should be established to facilitate supply-demand matching and information exchange among enterprises. These platforms can offer services such as market information, technical support, and partner matching, enhancing the ability of small and medium-sized cities to access resources from core cities and strengthening the borrowed size effect.

Second, expand the radiating effects of metropolitan areas and improve transportation infrastructure construction. The results of the heterogeneity tests show that metropolitan area development and transportation infrastructure are crucial for enhancing the borrowed size effect. First, policymakers should accelerate the integration of transportation infrastructure within metropolitan areas to improve the accessibility of small and medium-sized cities. Second, investment in transportation infrastructure between small and medium-sized cities and central cities should be increased to enhance regional connectivity. Additionally, while advancing the construction of the G60 Science and Technology Innovation Corridor, efforts should be made to avoid excessive concentration of innovation resources in core cities, which could negatively impact the borrowed size effect of small and medium-sized cities.

Third, optimize capital and labor misallocation to improve the effectiveness of borrowed size. Mechanism tests reveal that capital and labor misallocation are significant factors constraining the borrowed size effect. On one hand, regional financial markets should be improved to enhance capital allocation efficiency. Financial innovation should be leveraged to support the development of enterprises in small and medium-sized cities, addressing issues such as financing difficulties and high costs. On the other hand, labor mobility mechanisms should be refined to alleviate structural labor mismatches. Governments should promote reforms to the household registration system, reducing institutional barriers to cross-city labor mobility and enabling talent to flow more freely within the Yangtze River Delta region.

CRediT authorship contribution statement

Weishu Zhao: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Zhe Wang:** Writing – review & editing, Resources, Conceptualization. **Jiani Li:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Social Science Fund of China (24CJY136).

Data availability

Data will be made available on request.

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